| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 8(c)(ii) | (0, 10) minimum with correct reason<br><br>AND<br>(3, 37) maximum with correct reason | **3** | Reasons could be e.g.<br>1 A reasonable sketch of a negative cubic<br><br>2 Correct use of 2nd derivative = –12(0) + 18 = 18,<br>18 > 0, so (0, 10) is a minimum oe.<br>2nd derivative = –12(3) + 18 = –18, –18 < 0 so (3, 37) is a maximum oe.<br><br>3 Evaluates correctly values of *y* on both sides of both correct stationary points<br><br>4 Finds gradient on each side of both correct stationary points.<br><br>**B2** for 1 correct with correct reason for that stationary point<br><br>or for **both** *x*-values correct and reasonable sketch of a negative cubic,<br><br>or for correct substitution and evaluation of **both** of *their x*-values into *their* second derivative<br><br>or substitution and evaluation for one *x*-value on both sides of **both** of *their* stationary points to find the gradients soi<br><br>or **M1** for showing [2nd derivative =] $-12x + 18$<br>or correct FT *their* 2nd derivative<br><br>or substitution and evaluation shown for one *x*-value on both sides of **one** of *their* stationary points to find the gradients soi<br><br>or for sketch of any negative cubic. |
| 9(a)(i) | 5 | **3** | **M2** for $\frac{(12800-8000)\times 100}{8000\times 12}$<br>or **M1** for $[12800-8000=]\frac{8000\times 12\times r}{100}$<br>or 400 seen<br><br>If 0 scored, **SC1** for answer 13.3 or 13.33… |
| 9(a)(ii) | 4[.0] or 3.99… | **3** | **M2** for $\sqrt[12]{\frac{12800}{8000}}$<br>or **M1** for $12800 = 8000\times k^{12}$ for any $k$ |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 9(b) | 9 nfww | **3** | **M2** for $260\,000 \times \left(1+\frac{1.8}{100}\right)^8$ oe evaluated to 4 sf or better<br>or $260\,000 \times \left(1+\frac{1.8}{100}\right)^9$ oe evaluated to 2 sf or better<br>or **M1** for [300 000 = ] $260\,000 \times \left(1+\frac{1.8}{100}\right)^n$ oe soi<br>(Accept any inequality sign in [300 000 = ]) |
| 10(a) | −2.5 −1.25 5.5 | **3** | **B1** for each |
| 10(b) | Correct graph | **4** | **B3FT** for 8 or 9 correct points<br>or **B2FT** for 6 or 7 correct points<br>or **B1FT** for 4 or 5 correct points |
| 10(c) | $y$ = 2 drawn | **M1** | |
| | −2.75 to –2.65<br>–1.1 to −1.05<br>0.75 to 0.85 | **A2** | **A1** for 1 solution |
| 10(d) | –2.5 5.5 | **2** | **B1** for each |
| 11(a)(i) | −3.5 oe | **2** | **M1** for $g\left(\frac{1}{2}\right)$ seen or $3\left(\frac{1}{x}\right)-5$ or better |
| 11(a)(ii) | $\frac{x+5}{3}$ oe final answer | **2** | **M1** for correct first step $y+5=3x$, $\frac{y}{3}=x-\frac{5}{3}$ or $x=3y-5$ |
| 11(b) | $3x-11$ final answer | **2** | **M1** for $3(x-2)-5$ |
| 11(c)(i) | 5 | **2** | **M1** for $\frac{1}{3x-5}[=0.1]$ |
| 11(c)(ii) | 4 nfww | **2** | **M1** for $2^x-(3\times7-5)[=0]$ or better |